

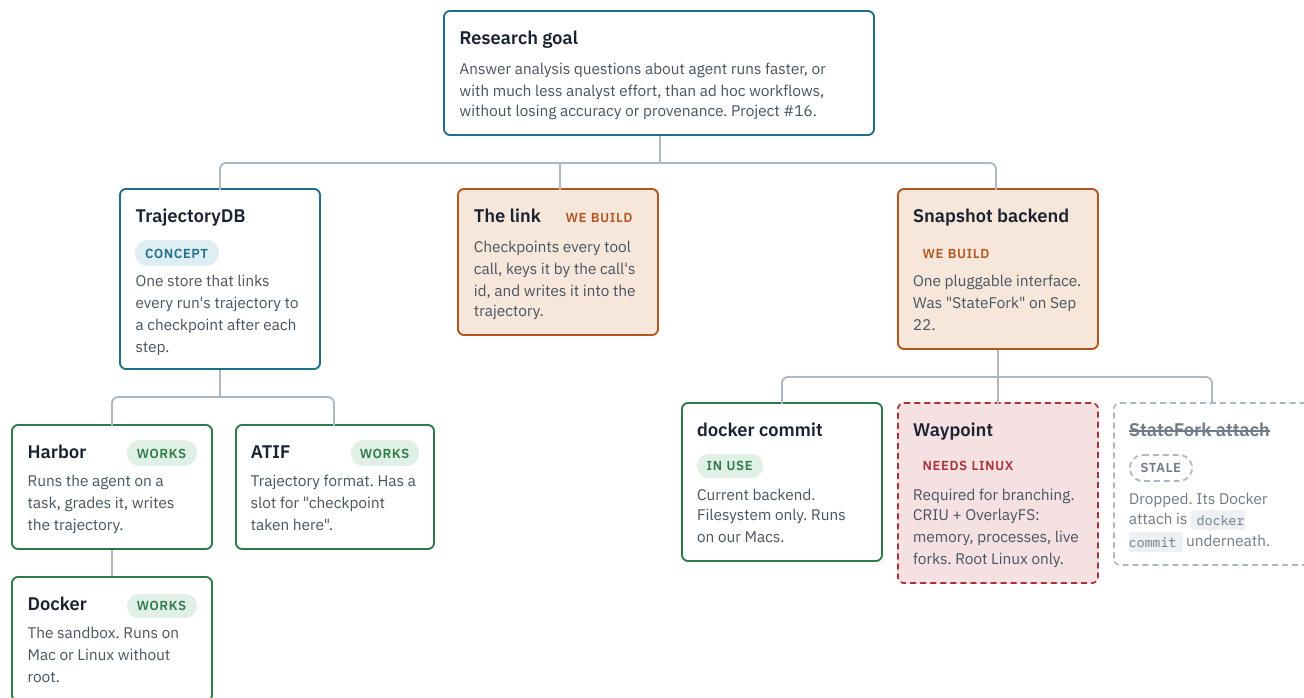
Branching needs a Linux server

BOTTOM LINE

The project brief requires branchable environment state alongside agent trajectories. Only Waypoint branches a live sandbox, and it runs only on a root Linux host. Our Macs cannot meet the brief, so we need a Linux server.

Revised architecture

This is the same tree as Sep 22 with the state half corrected. StateFork is the common interface and benchmark layer. The CRIU engine that captures memory and forks live sandboxes is Waypoint. StateFork's Docker path is `docker commit`, the same thing we already run, so we dropped it (ADR-0004).



□ Concept □ Works today on macOS □ Implemented by us □ Blocked without a Linux host

□ Stale since Sep 22

Three blockers

We checked this against the StateFork and Waypoint source (v0.7.0, vendored under `vendor/`) and the Harbor source. A Linux server removes the first blocker. The second is integration work we can do without patching Harbor. The third rules out every shortcut.

- **It needs root on a Linux host.** Waypoint is a Go binary built on CRIU and OverlayFS, and it requires root. On macOS, Docker Desktop's VM has no CRIU at all (docker/for-mac#1059, open since 2017). OrbStack exposes `docker checkpoint`, but only for arm64-native containers, and it is still not a Linux host we control.
- **It does not attach to Harbor's containers.** Harbor starts every task with `docker compose`. Waypoint stages its own rootfs and PTY sessions, and `waypoint_attach` takes a Waypoint session id, not a container or a PID. The way in is a custom Harbor environment, loaded with `--env module:Class`, that builds each task with `waypoint build` from the task's Dockerfile and runs commands with `waypoint exec`. Harbor stays unpatched. Open risk: tasks that need several compose services or network isolation.
- **The other CRIU routes are dead ends.** StateFork's `docker_attach` is `docker commit` (`container_env_manager.py:120`). `hybrid_attach` captures memory, but only on Podman with runc and root, so Harbor runs would have to move off Docker. Docker's own `docker checkpoint` is still experimental and fails on containers with an external terminal, which is how agents run.

CRIU is also expensive at our granularity. Each dump is roughly as large as the process memory (about 1.7 s per 2 GB, arXiv:2510.05556), and we would take one for every state-changing tool call in every trial.

What Harbor and Docker still record

RECORDED ON A MAC TODAY

- The full ATIF trajectory for every trial (Harbor)
- Grade, tokens, and cost per trial
- An environment checkpoint after every tool call: hook, then watcher, then `docker commit`
- The join: each checkpoint keyed by `tool_call_id` and inserted into the trajectory
- Each checkpoint is a runnable image, so `docker run` shows the filesystem at step N

LOST WITHOUT WAYPOINT

- Memory and running processes (servers, daemons)
- Shell state: working directory, exported variables, background jobs
- Native live branching: forking a running sandbox mid-trial into concurrent copies
- Exact restore: a commit image restarts cold, with a fresh shell and no processes

What dropping Waypoint costs

DIMENSION	DOCKER COMMIT (NOW)	WAYPOINT (CRIU)	COST TO US
Filesystem state	Captured	Captured	None
Memory, processes, shell	Lost	Captured	Low. Terminal-Bench grades files, and our analysis questions are framed around filesystem changes. Most shell side effects that matter end up on disk.
Branching	Cold only: start a new container from the step-N image	Concurrent live forks from any checkpoint	Crucial. The project brief asks for branchable environment state. <code>docker commit</code> alone does not meet it.
Host	Mac or Linux without root	Root Linux only	Blocks our Macs. None on a server.
Harbor integration	Attaches to its containers as they are	Needs a custom Harbor environment	Medium. Integration work. Harbor stays unpatched.
Storage per checkpoint	Re-tars the writable layer; commits do not dedupe	Memory-sized dump, about 1.7 s per 2 GB	Low for Terminal-Bench-sized write sets

Dropping Waypoint fails the brief. ADR-0004 needs to be reopened. `SnapshotBackend` is pluggable, so the capture pipeline carries over, with a new `corpus_id`.

Terminal-Bench on Apple Silicon

This is a second, independent reason to get a Linux server. Harbor does not pin a platform, so on our M-series Macs the containers run as arm64. Some Terminal-Bench tasks are x86-only. The upstream tracker lists these as of Sep 23:

TASK	ISSUE	FAILURE ON ARM64
fix-uautomizer-soundness	#1601	Silent: all verdicts UNKNOWN, even the reference solution scores 0. Looks like an ordinary agent failure.
coq-block-bound	#1802	Image not multi-platform

TASK	ISSUE	FAILURE ON ARM64
hof-topology-interpenetration	#1803	Image not multi-platform
math-eval-grader	#1804	Image not multi-platform
gsea-proteomics	#1805	Image not multi-platform
wdm-design	#1806	Image not multi-platform

- **The list is incomplete.** Cross-platform CI is only proposed (#1793). Because #1601 fails silently, other broken tasks can hide in a Mac-recorded corpus as fake agent failures, and we cannot measure that bias.
- **Emulation is for development only.** `DOCKER_DEFAULT_PLATFORM=linux/amd64 harbor run ...` runs x86 images under emulation, and the #1601 reference solution then scores 1. Emulation distorts timing (checkpoint latency, task timeouts), so emulated runs cannot go into a recorded corpus.

Our macOS run

PENDING

Shawn is running the subset natively on arm64, then again under amd64 emulation. This table gets filled in before Friday.

MEASURE	ARM64 NATIVE	AMD64 EMULATED
Tasks run	-	-
Fail to build or start	-	-
Reference solution scores 0	-	-
Tasks newly broken, not in the tracker	-	-

Request a Linux server on Friday

One Linux machine fixes both problems. Root lets Waypoint branch live sandboxes, which the brief requires. x86_64 removes the architecture bias.

QUESTION FOR THE PROFESSOR

Can the course or department provide or fund an x86_64 Linux server or VM with root access to record the corpus?

- x86_64, with root, running Docker Engine (not Docker Desktop)
- Enough disk for one committed image per tool call in every trial
- Can run long batch jobs unattended
- A kernel that supports CRIU, which Waypoint needs for branching
- Disk for CRIU dumps, which grow with process memory

Development stays on our Macs until the server arrives, with emulation for x86-only tasks. The server blocks branching and the recorded corpus, not the Sep 25 baseline (stock Harbor, no checkpoints). Team decision for Friday: reopen ADR-0004 and plan the Waypoint environment.

Sources: docs/project-brief.md, docs/decisions/0004-filesystem-only-checkpoints.md, docs/research/2026-09-23_checkpoint-platform-research.md, vendor/statefork and vendor/waypoint at v0.7.0, arXiv:2510.05556, docker/for-mac#1059, harbor-framework/terminal-bench issues #1601, #1793, #1802-#1806. Supersedes the StateFork section of the Sep 22 brief. State as of Sep 23, 2026.